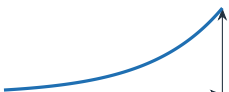
# Условие

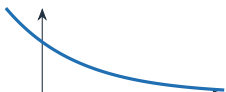
Нека $X$ има плътност $f_X(x) = \lambda e^{-\lambda x}\mathbb{1}_{(0,\infty)}(x)$. Намерете разпределенията на (а) $Y = -X$; (б) $Y = 2X - 1$; (в) $Y = \sqrt{X}$.

# Формула за смяна на променливите

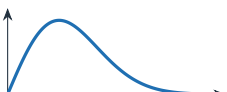
$f_{g(X)}(y) = f_X(g^{-1}(y))\,|(g^{-1}(y))'|$. Нека $\lambda = 1$ за картинката:



(а) $Y = -X$



(б) $Y = 2X - 1$



(в) $Y = \sqrt{X}$

**Отговор.** (а) $g^{-1}(y) = -y$: $f_Y(y) = \lambda e^{\lambda y} \mathbb{1}_{(-\infty,0)}(y)$;
(б) $g^{-1}(y) = \frac{y+1}{2}$: $f_Y(y) = \frac{\lambda}{2} e^{-\lambda \frac{y+1}{2}} \mathbb{1}_{(-1,\infty)}(y)$;
(в) $g^{-1}(y) = y^2$: $f_Y(y) = 2\lambda y e^{-\lambda y^2} \mathbb{1}_{(0,\infty)}(y)$.